AI Job Market & Student AI Usage Analytics Report (2020–2025)

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1. Installation and Setup

1.1 Software and Tools Used

- Microsoft Power BI Desktop: For interactive dashboards and advanced DAX-based analytics.
- Excel/CSV Files: Source data loaded directly from structured `.csv` and `.xlsx` formats.
- DAX (Data Analysis Expressions): For custom measures and KPIs.

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2. Data Preparation and Transformation

2.1 Dataset Overview

Dataset	Description	Rows	Format
ai_job_dataset.csv	AI job market data including salaries, remote %, company size	~5,000	CSV
ai_job_dataset1.csv	Extension of job dataset with job title & industry detail	~5,000	CSV
salaries.csv	Historical salary trends by job title & company size	~1,500	CSV
Students.csv	AI tool usage by college students across streams and years	~3,600	CSV
AI On-Campus Survey.xlsx	AI awareness, ChatGPT usage, school use breakdown	~258	Excel

2.2 Key Transformations

- Standardized column headers.
- Normalized company size, experience labels (EN, MI, EX).
- Derived Month-Year field from timestamps:
  `MonthYear = FORMAT ('Form Responses 1'[Timestamp], "YYYY-MM")`
- Merged experience and company size fields for cohesive visualizations.
- Converted Likert scale responses to numeric format where applicable.

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3. Power BI Measures and DAX Calculations

3.1 AI Usage Among Students

- % Knows ChatGPT:

```
DIVIDE (
    CALCULATE (COUNTROWS ('Form Responses 1'), 'Form Responses 1'[Do you know
what Chat-GPT is?] ="Yes"),
    COUNTROWS ('Form Responses 1')
) * 100
```

- Avg Career Interest in AI: `AVERAGE('Form Responses 1'[Career Interest in AI])`
- Avg Personal Use of AI: `AVERAGE('Form Responses 1'[Personal Use of AI])`
- Avg Work Use of AI: `AVERAGE('Form Responses 1'[Work Use of AI])`

3.2 AI Job Market Measures

- Total Employees: `COUNTROWS('ai_job_dataset')`
- Average Salary: `AVERAGE('ai_job_dataset'[salary_in_usd])`
- Avg % Remote Work: `AVERAGE('ai_job_dataset'[remote_ratio])`
- Salary by Title/Experience: `AVERAGE('ai_job_dataset'[salary_in_usd])` grouped by job_title and experience_level

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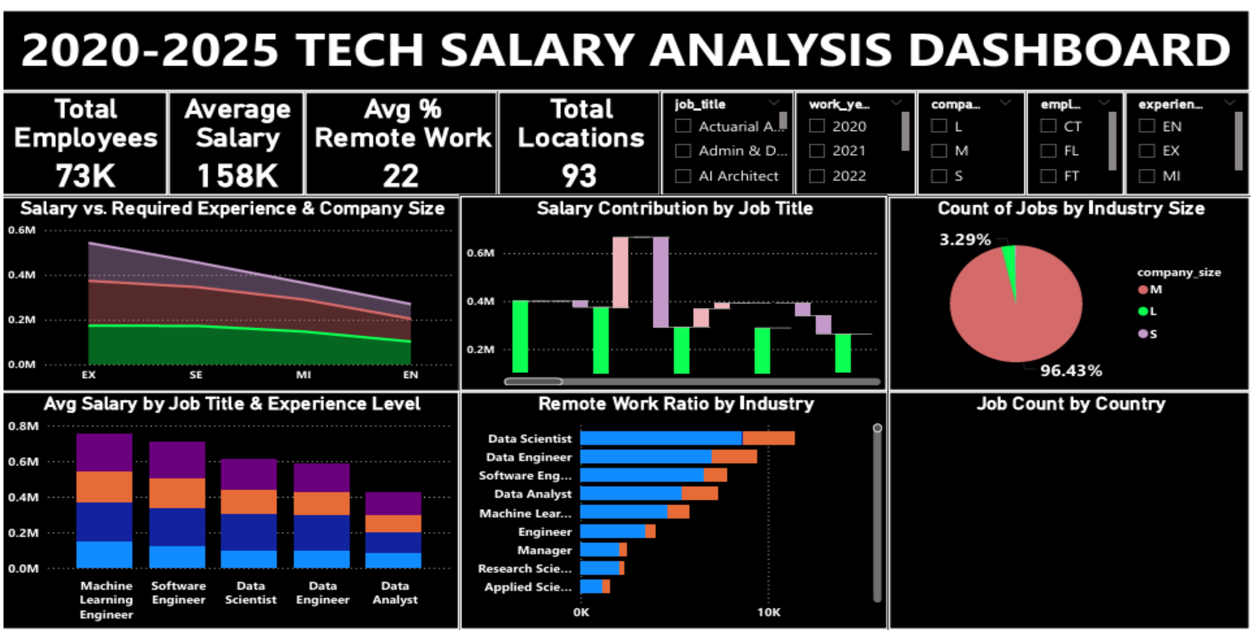
4. Power BI Dashboard Analysis

Dashboard 1: 2020–2025 Tech Salary Analysis



- **Total Employees**: 73,000
- **Average Salary**: $158,000
- **Avg % Remote Work**: 22%
- **Total Locations**: 93

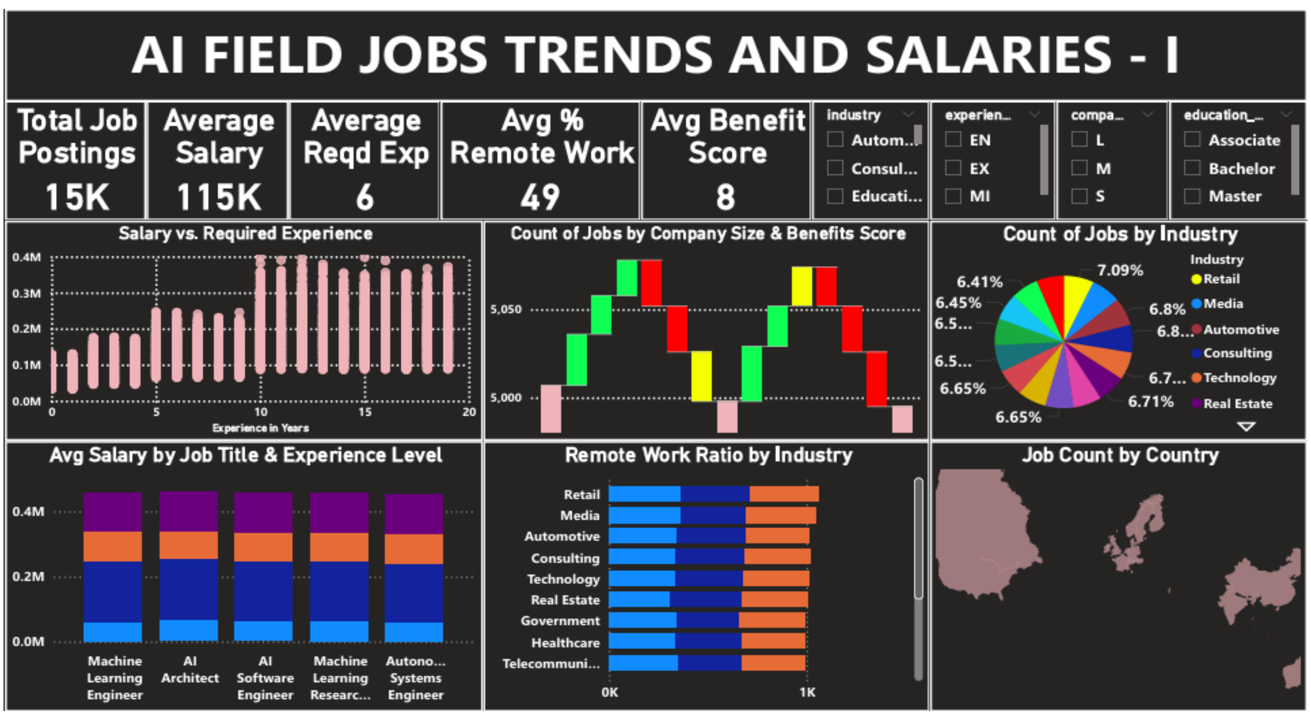
Detailed Analysis:

- Employees are highly concentrated in large firms with expansive global reach.
- Salary distribution is skewed toward high-experience roles in large companies.
- Job titles such as Machine Learning Engineer, Software Engineerpo and Data Scientist receive top-tier compensation.
- Remote work adoption is limited but growing, with certain industries lagging behind.
- Visuals suggest geographic clustering in North America and Western Europe.

Recommendations:

1. Encourage mid-sized companies to adopt hybrid work models.
2. Promote salary benchmarking practices for startups to remain competitive.
3. Explore relocation support for employees in high-cost regions.
4. Improve DEI (**Diversity, Equity, and Inclusion**) representation across international branches.
5. Use location intelligence to inform market expansion strategy.

Dashboard 2: AI Job Trends and Salaries - Part I



- **Total Job Postings**: 15,000

- **Average Salary**: $115,000
- **Avg Required Experience**: 6 years
- **Avg Benefit Score**: 8

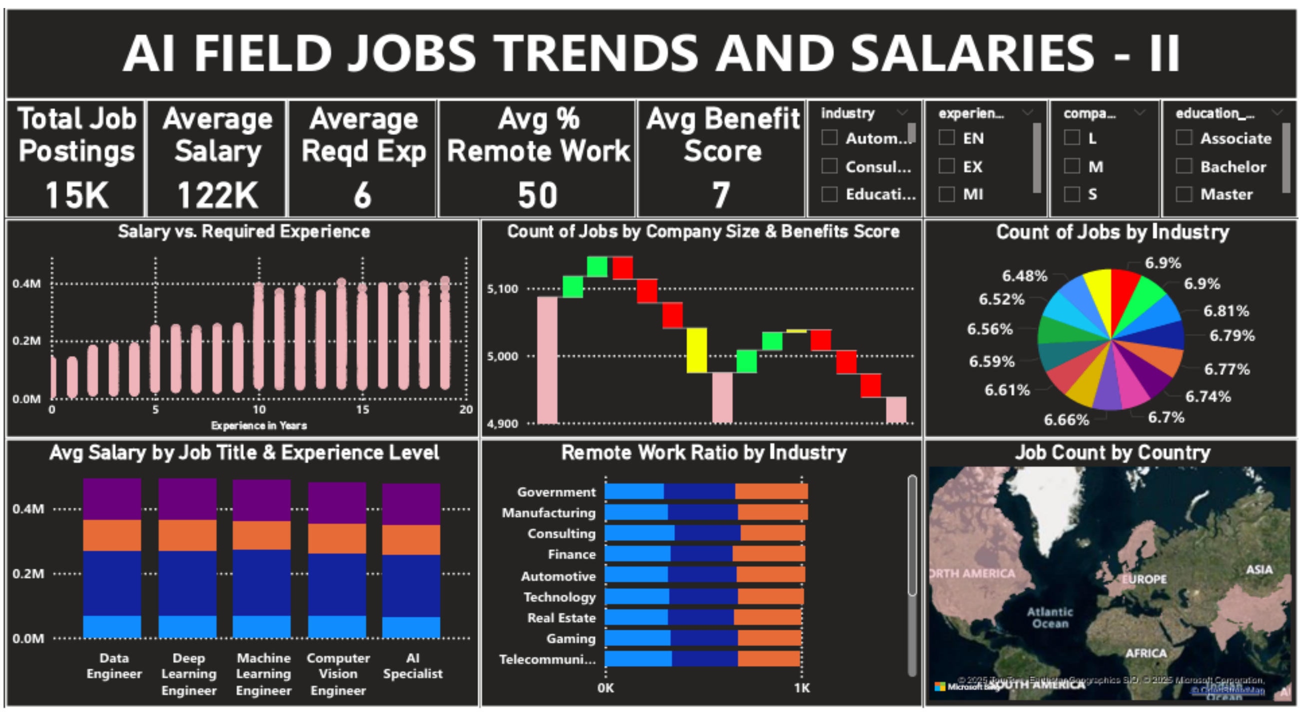
Detailed Analysis:

- Senior roles in AI software architecture and ML research dominate in compensation.
- Education level has a significant impact: Master's degree holders earn substantially more.
- Remote work is common in Tech, Media, and Consulting, but rare in Government and Healthcare.
- Experience level correlates directly with salary until mid-career (10 years).
- Certain job titles are over-represented in underpaid roles due to industry disparity.
- More job postings commonly in Retail, Media and Automotive Industry.

Recommendations:

1. Align job postings with appropriate experience brackets.
2. Invest in upskilling programs for junior AI developers.
3. Increase benefit score transparency in postings.
4. Leverage employer branding in remote-first roles.
5. Consider regional salary modifiers in job ads.

Dashboard 3: AI Job Trends and Salaries - Part II



- **Total Job Postings**: 15,000
- **Average Salary**: $122,000

- **Avg Required Experience**: 6 years
- **Avg Benefit Score:** 7

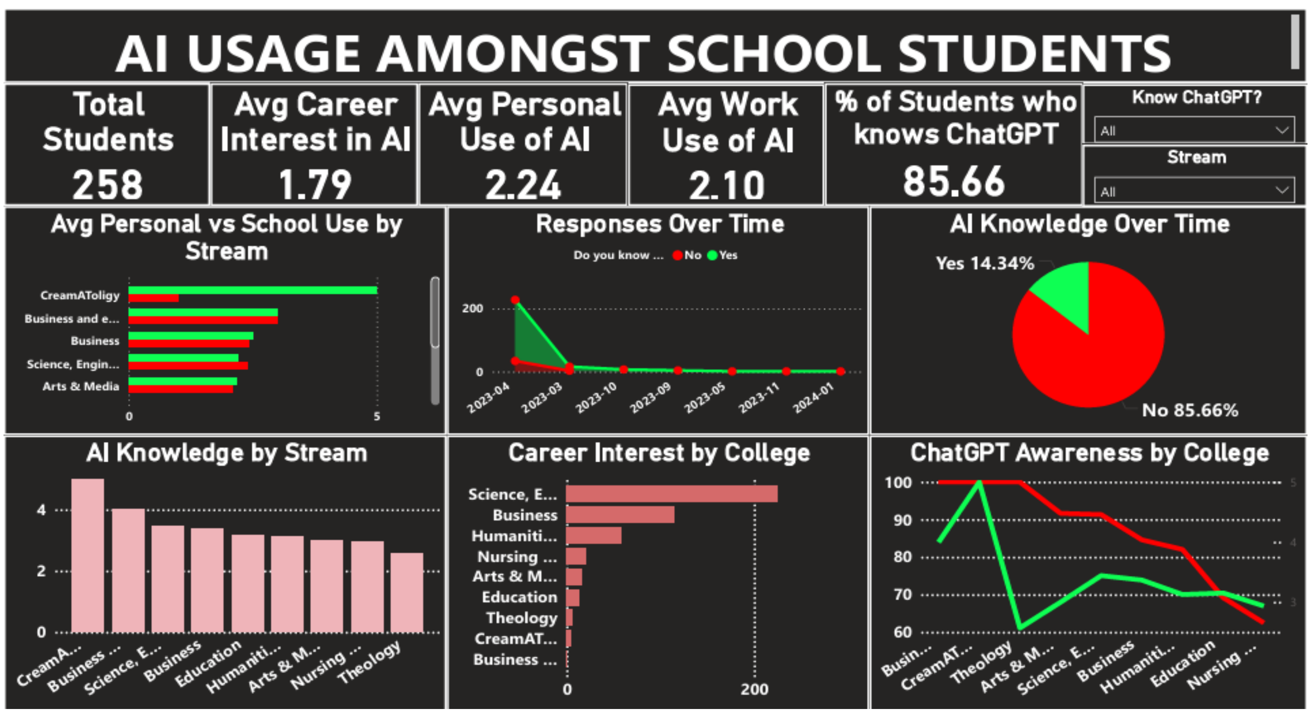
Detailed Analysis:

- Deep Learning and Computer Vision specialists have salary premiums.
- Real Estate and Gaming industries lag in AI salary competitiveness.
- There is diminishing return on salary beyond 15 years' experience.
- Company size positively correlates with both salary and benefit score.
- Salary bands are more consistent in Technology and Consulting sectors.

Recommendations:

1. Standardize job titles and expectations across industries.
2. Highlight long-term growth potential in niche AI domains.
3. Use career mapping tools to retain mid-career AI talent.
4. Adjust hiring strategy based on experience salary curves.
5. Increase collaboration between HR and Data Science units.

Dashboard 4: AI On-Campus Usage Report (Survey-Based)



- **Total Students Surveyed**: 258
- **% Know ChatGPT**: ~85.66%
- **Avg Career Interest in AI**: 1.79
- **Avg Personal Use**: 2.24
- **Avg Work Use**: 2.10

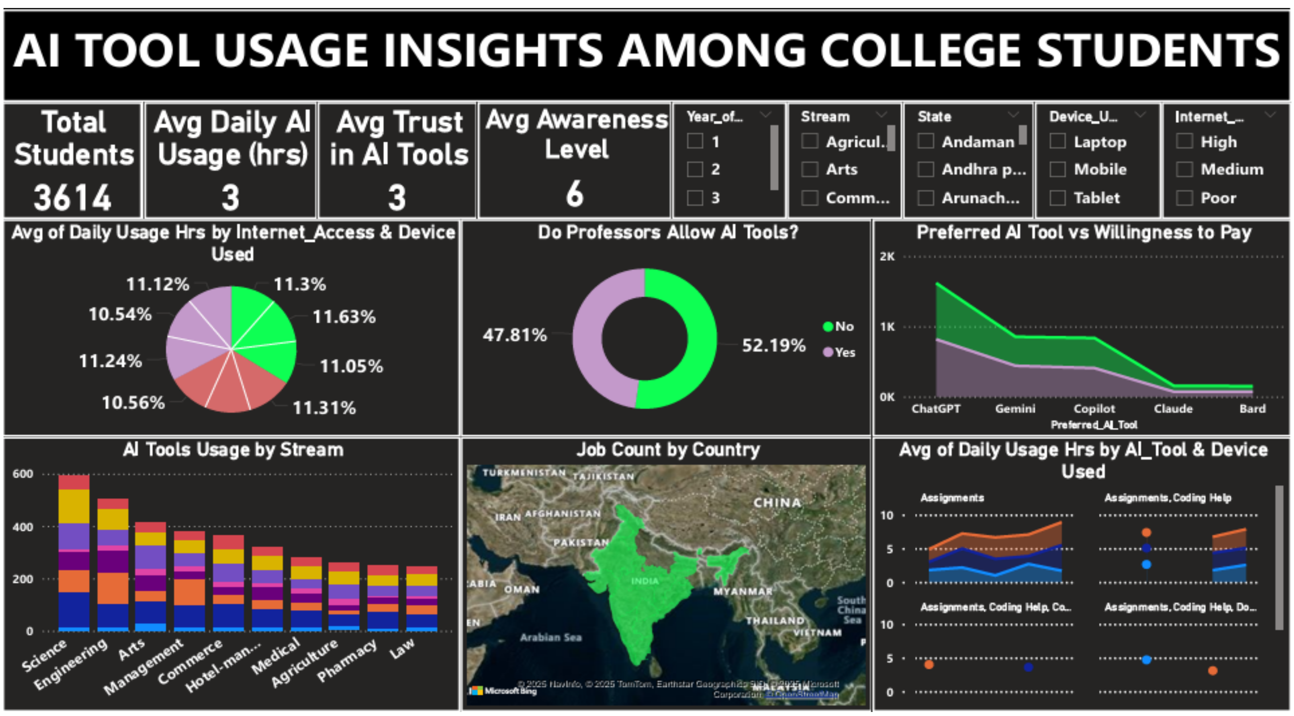
Detailed Analysis:

- Awareness is high, but practical usage remains moderate.
- Engineering and Business students show highest enthusiasm for AI Fields.
- Streams like Theology and Arts report significantly lower AI integration.
- Temporal analysis indicates seasonal spikes in AI tool exploration.
- Positive linear correlation exists between awareness and usage frequency.

Recommendations:

1. Introduce early AI literacy workshops in all streams.
2. Track usage evolution through semester-based reporting.
3. Integrate AI tools in assignment frameworks.
4. Establish incentives for faculty to promote AI adoption.
5. Develop AI career mentoring programs.

Dashboard 5: AI Tool Usage Insights Among College Students



- **Total Students Surveyed**: 3,614
- **Avg Daily AI Usage**: 3 hours
- **Avg Trust in AI Tools**: 3/5
- **Avg Awareness Level**: 6/10

Detailed Analysis:

- Most popular tools are ChatGPT, Gemini, Copilot.

- AI usage is higher among laptop users with stable internet access.
- High AI engagement reported in Science, Engineering, and Law.
- 52% of professors do not allow AI tools, limiting usage context.
- Students using AI for assignments and coding show longer daily engagement.

Recommendations:

1. Provide subsidized devices to digitally underserved students.
2. Conduct faculty-AI alignment training.
3. Implement ethical AI use guidelines at institution level.
4. Track tool usage trends per department.
5. Launch cross-stream AI hackathons to boost innovation.

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6. Overall Strategic Recommendations

1. **Bridge Academic-Industry Gap**: Develop structured pipelines between AI education and employment.
2. **Standardize Salary Frameworks**: Industry-wide benchmarking for roles like ML Engineer and Data Scientist.
3. **Expand Remote Work Policies**: Especially in Tech and Consulting industries where adoption has proven ROI.
4. **Equip Campuses with AI Infrastructure**: Devices, access, software licenses, and AI content libraries.
5. **Implement Inclusive AI Literacy Programs**: Across all academic streams, including underrepresented fields.

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7. Conclusion

This analysis highlights the dual transformation occurring in AI: professional landscapes are evolving rapidly with growing demand and differentiated compensation structures, while academic environments are beginning to respond with increased student interest and engagement. Bridging these two trends through coordinated upskilling, policy support, and platform access will be key to ensuring a sustainable and inclusive AI-driven future.